ECOENERGY CONSULTANCY

Desktop Terrain Screening

Kota Setar paddy, Lot A

Kota Setar, Kedah

SCREENING GRADE — NOT A SURVEY

Report ID	BEq_flm-piWOPpdDonFm2w
Generated	13 October 2026, 09:36 MYT
Site file	flat_site_kedah.kml
Scheme	Large Scale Solar (LSS)
Project stage	Exploring options
Centroid	6.04221° N, 100.35545° E

About this report

This desktop screening summarises the terrain inside the boundaries supplied in flat_site_kedah.kml, using the Copernicus DEM GLO-30. It is meant for comparing candidate sites at an early stage and for deciding which surveys and studies to commission next.

It reports the area, elevation and slope of each site, the share of each site in the KPKT hill-land slope classes, rule-based findings and recommended follow-up studies. Every finding is generated automatically from the elevation data by fixed rules; nobody has visited the site.

Read the caveats and the disclaimer at the end of the report before relying on any figure in it.

At a glance

Location	Kota Setar, Kedah
Total site area	42.16 ha
Features analysed	1 site · 1 route (2.03 km) · 1 point
Elevation range	1.2–5.4 m (EGM2008)
Overall KPKT split	Class I 100.0% · Class II 0.0% · Class III 0.0% · Class IV 0.0%
Maximum slope	3.9° (in “Lot A paddy”)
Findings	0 warnings · 0 cautions · 3 for information
Elevation model (DEM)	Copernicus DEM GLO-30 · DSM (surface model: includes tree canopy and buildings) · 30 m nominal, analysed at 30.7 m

Findings

Each finding below is computed by fixed rules from the elevation data and is listed most severe first: warnings, then cautions, then information.

INFORMATION

Site area screens as KPKT Class I (below 15°)

No site has more than 0.5% of its area at 15° or steeper on this DEM, so the land screens as Class I, which the KPKT hill-land guidelines generally treat as developable. A 30 m elevation model smooths out short steep features such as bund faces, drain banks and cuttings, so a survey can still find locally steeper ground.

Source: KPKT hill-land development guidelines (Garis Panduan Perancangan Pembangunan di Kawasan Bukit dan Tanah Tinggi)

INFORMATION

Route gradients

Steepest gradient along each route, measured over at least 90 m — “33 kV cable route”: max grade 2.1% over 2.03 km (total climb 5 m). Short pitches at drains, embankments and road crossings can be steeper than a 30 m DEM shows.

INFORMATION

Check EIA requirements for the total area

Total polygon area is 42.16 ha. Whether an Environmental Impact Assessment is required depends on the activity and area categories in the schedules of the Environmental Quality (Prescribed Activities) (Environmental Impact Assessment) Order 2015, and on factors such as hill land, forest clearing or conversion of agricultural land. Have a qualified EIA consultant check the site against the current schedules.

Source: Environmental Quality (Prescribed Activities) (Environmental Impact Assessment) Order 2015, under the Environmental Quality Act 1974

Site statistics

Sites

Site	Area (ha)	Elevation min–max (m)	Relief (m)	Slope (°)			KPKT class (% of site area)			
				Mean	P90	Max	I	II	III	IV
Lot A paddy	42.16	1.2–5.4	4.2	0.7	1.6	3.9	100.0	0.0	0.0	0.0

Elevations are DEM surface heights in metres above EGM2008. P90 is the 90th-percentile slope: nine-tenths of the site area is at or below it. KPKT classes: I below 15°, II 15–25°, III 25–35°, IV 35° and steeper.

Slope distribution

Site	Share of site area in each slope band (%)						
	0–2°	2–5°	5–10°	10–15°	15–25°	25–35°	>35°
Lot A paddy	92.6	7.4	0.0	0.0	0.0	0.0	0.0

Routes

Route	Length (km)	Elevation min–max (m)	Total ascent (m)	Max grade (%)
33 kV cable route	2.03	2.2–5.5	5.3	2.1

Points

Point	Latitude (°)	Longitude (°)	Elevation (m)
PMU	6.05290	100.37380	5.3

Figures

Figure 1. Site layout

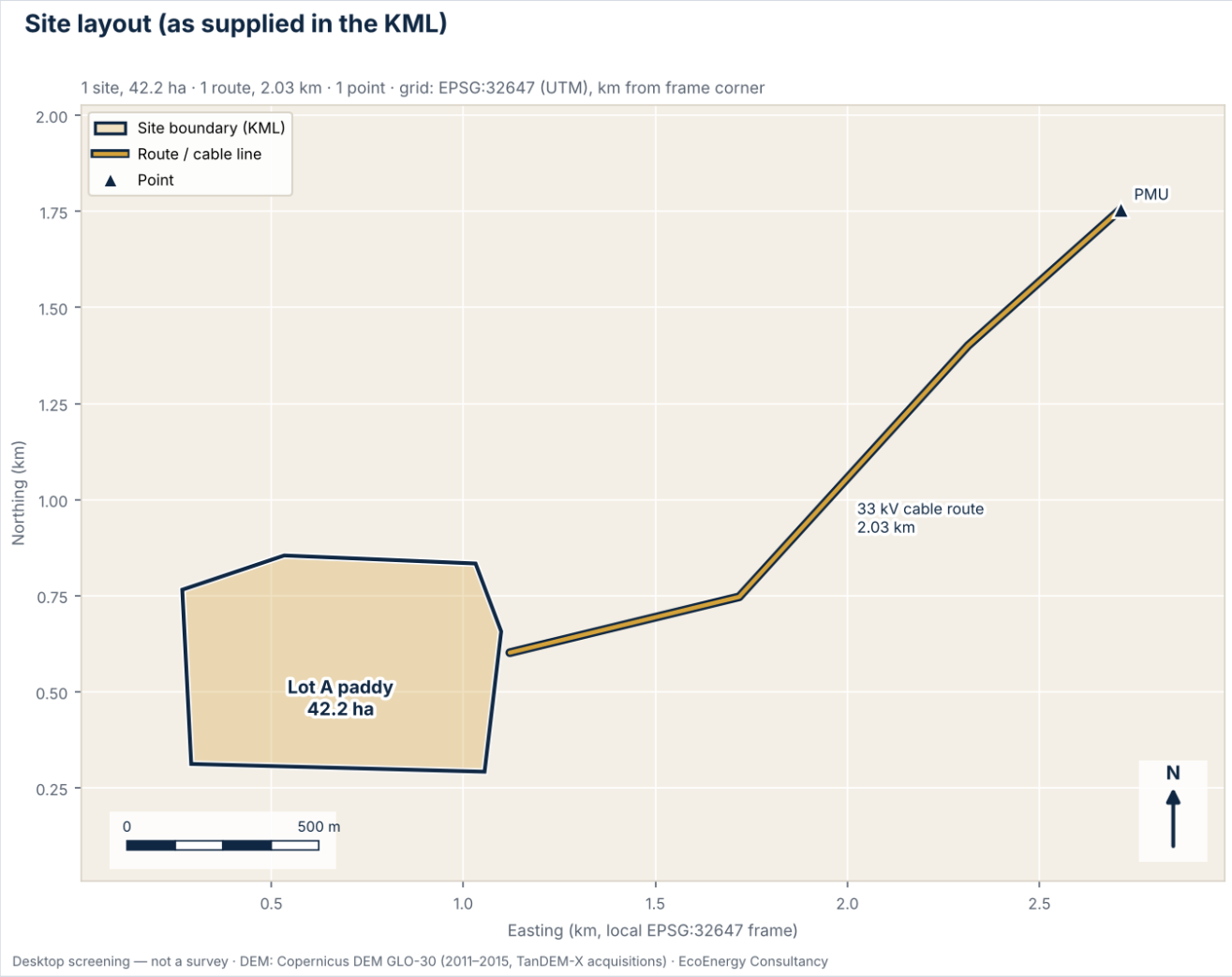


Figure 1. Site boundaries, routes and points as supplied in the KML, with areas in hectares, on a local UTM grid (EPSG:32647).

Figure 2. Elevation and contours



Figure 2. Shaded Copernicus DEM GLO-30 surface elevation with contours every 5 m; tree canopy and buildings are included in the surface.

Figure 3. Slope classes

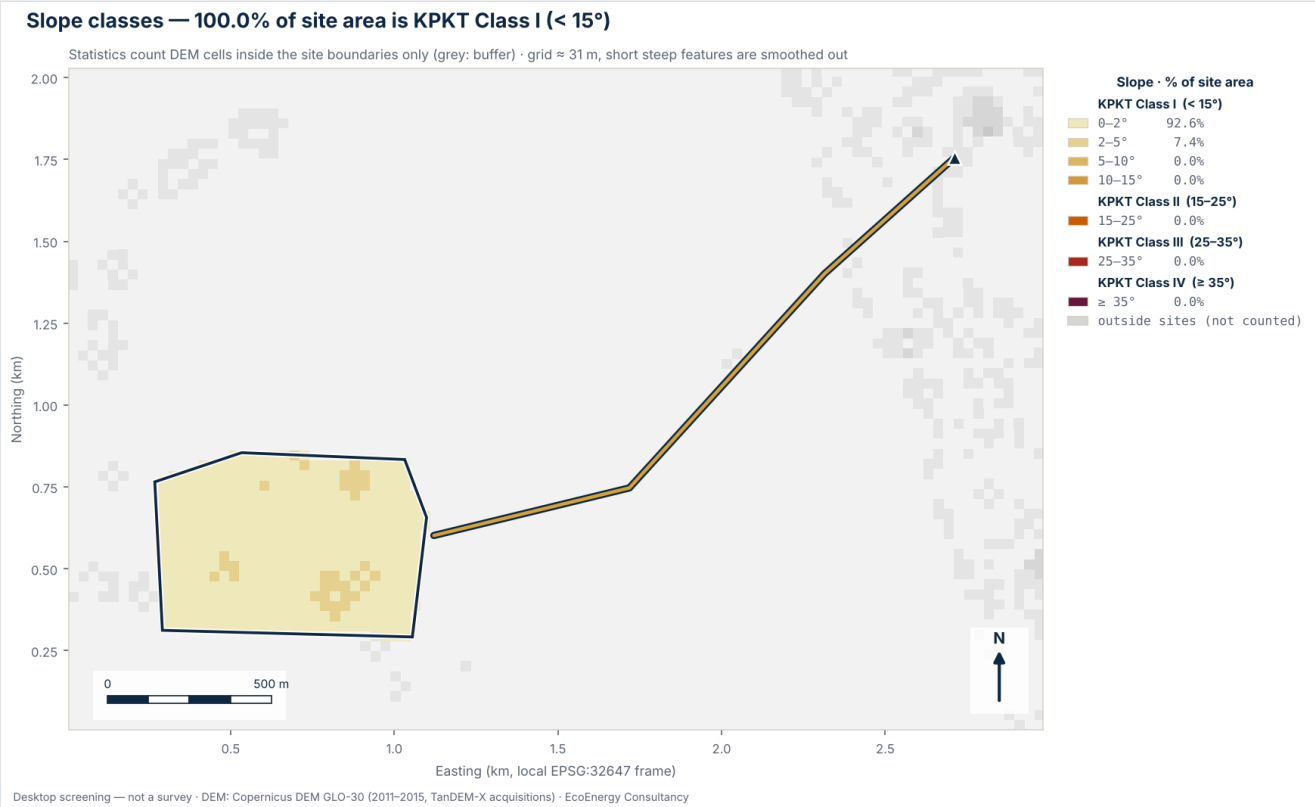


Figure 3. Slope from the elevation model in seven bands, with the KPKT class boundaries at 15°, 25° and 35°; 100.0% of the site area is KPKT Class I (below 15°).

Figure 4. Terrain profiles

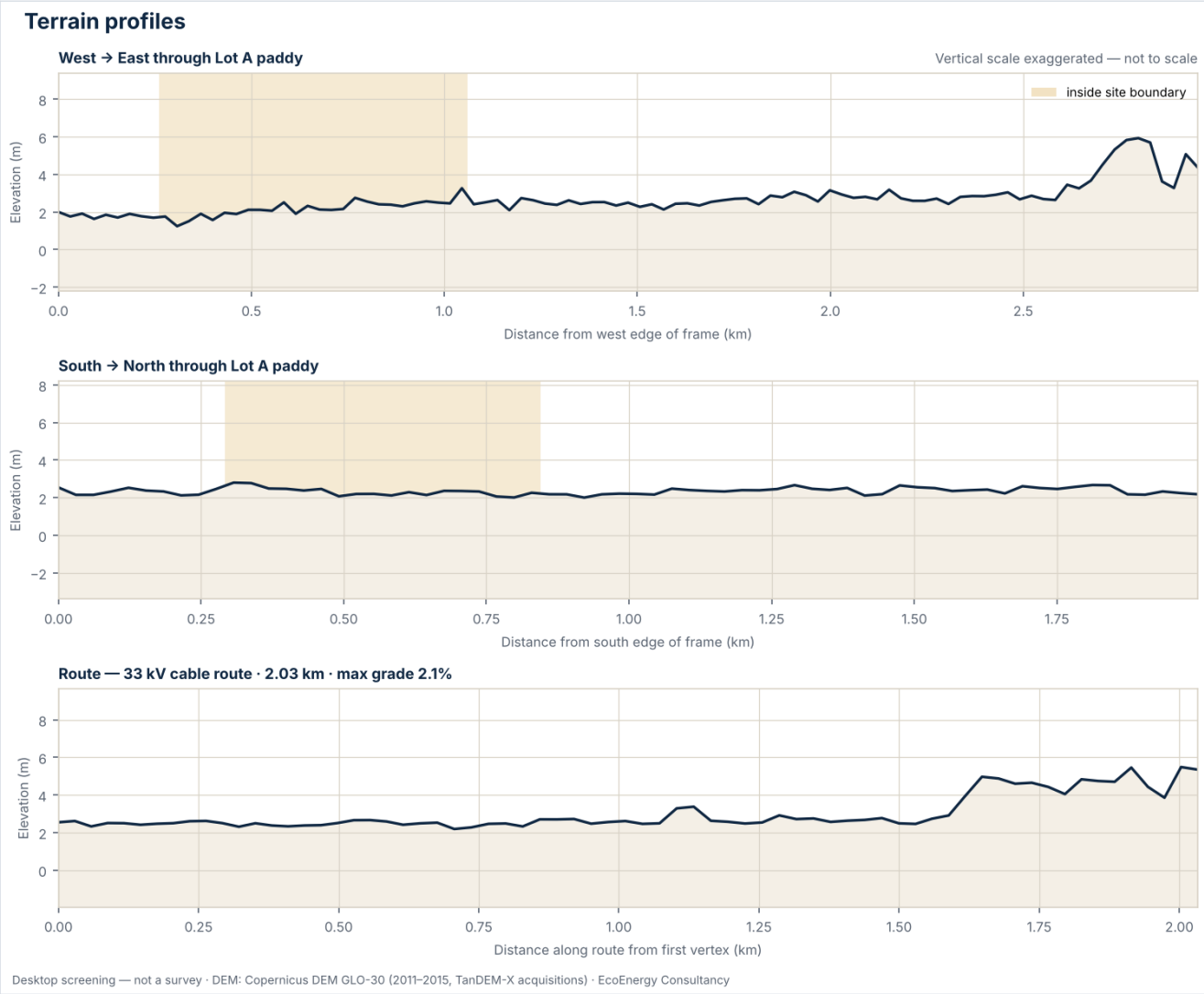


Figure 4. West–east and south–north sections through “Lot A paddy”, plus the elevation profile along 1 route; the vertical scale is exaggerated and not to scale.

Recommended follow-up studies

1. **Licensed topographic survey**
   Before layout, earthworks or drainage design, commission a topographic survey by a licensed land surveyor (Licensed Land Surveyors Act 1958), or LiDAR / drone photogrammetry tied to surveyed ground control points.
2. **Geological map review (JMG)**
   Review the Jabatan Mineral dan Geosains (JMG) geological map sheet for the site: formation, lithology and any karst or ex-mining ground that affects piling, anchoring or foundations.
3. **Rainfall and climate data (MET Malaysia)**
   Obtain climate normals and rainfall records from the nearest MET Malaysia station; rainfall intensity feeds stormwater design to MSMA 2nd Edition.
4. **Surrounding land-use scan (1 / 5 / 10 km)**
   Map settlements, schools, rivers and water intakes, forest reserves, industry and mining within 1 km, 5 km and 10 km of the site as context for EIA and SIA scoping.
5. **Land title, zoning and local authority requirements**
   Confirm the land title, category of land use and any express conditions, the zoning in the local plan (Rancangan Tempatan), and the requirements of the local authority (PBT) for this type of development.
6. **Route wayleave and crossings**
   Confirm land access and wayleaves along the cable or access route and identify road, river and railway crossings and TNB technical requirements; none of these show on an elevation model.

Caveats and limitations

- Screening grade only: this is a desktop terrain screening from an open global elevation model, not a survey.
- The Copernicus DEM GLO-30 is a surface model: it records the top of tree canopy and buildings, not bare ground. Plantations, forest and built-up areas read higher than the ground beneath them, and slopes at their edges can be exaggerated.
- Elevations show the ground as it was in 2011–2015 (TanDEM-X acquisitions). Ponds, cuttings, platforms and buildings made since then are not represented.
- The nominal 30 m resolution (analysed at about 31 m) smooths out bunds, drains and short steep cuttings, so local slopes are underestimated.
- Site geometry is used exactly as supplied in the uploaded KML/KMZ file; it has not been checked against title plans or survey boundaries.
- Not for authority submissions or design: planning applications, earthworks, drainage and structural design need a licensed survey and qualified engineers.

Data sources and attribution

- **Elevation data:** Copernicus DEM GLO-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA; all rights reserved.
- **Elevation model:** Copernicus DEM GLO-30; DSM (surface model: includes tree canopy and buildings); acquired 2011–2015 (TanDEM-X acquisitions); vertical datum EGM2008; 30 m nominal resolution, analysed at 30.7 m over 6,402 cells (0.0% without data).
- **Administrative boundaries:** geoBoundaries (gbOpen) — states © OpenStreetMap contributors (ODbL 1.0); districts citypopulation.de (CC BY 3.0).
- **Projection:** areas, lengths and slopes are computed on a local UTM grid (EPSG:32647, WGS 84).

- **Site geometry:** flat_site_kedah.kml, used as supplied (SHA-256 045a5befe1ca93a2…).
- **Analysis:** EcoEnergy terrain screening engine v1

Disclaimer

This report is a desktop screening produced automatically from open global elevation data, for early-stage site comparison only. It is not a topographic survey, not engineering design and not for submission to any authority. A topographic survey by a licensed land surveyor (Licensed Land Surveyors Act 1958) is required for design. EcoEnergy Consultancy accepts no liability for decisions made solely on the basis of this screening.

Next step

Book a free 30-minute scheme-fit consult

Walk through these results with an EcoEnergy consultant. In 30 minutes we will go over the findings, check which scheme the site fits, and agree which surveys and studies to commission first.

Book your consult »

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